

Effect of lipid-based suspension of *Epimedium koreanum* Nakai extract on sexual behavior in rats

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Abstract

Ethnopharmacological relevance: Herba of *Epimedium koreanum* is used in traditional Chinese and Korean herbal medicine as a potent enhancer of erectile function. Icaritin, the main active component of *Epimedium koreanum*, possesses many biological effects, such as improving cardiovascular function, hormone regulation, immunological function modulation, and anti-tumor activity.

Aim of the Study: This study supports the traditional use of extracts from *Epimedium* species in erectile dysfunction.

Materials and Methods: The *Epimedium koreanum* dry extract was suspended in wheat germ oil using lecithin and bee wax for oral administration. The effect of oral administration of two compositions (E-01 and E-02) standardized by their icaritin content on the number of complete intromissions, the number of ejaculations, and the latent period of ejaculation (LPE) in rats were evaluated. E-01 and E-02 were administered orally for 10 days to the experimental animals. The control animals received olive oil for 10 days. On day 10, 0.5 h after the dose was administered to male rats, one virgin female rat was placed with one male rat.

Results: The number of complete intromissions increased to 23.3 ± 2.6 in the E-01 and E-02 group (dose 300 mg/kg body weight) (b.wt) and to 20.1 ± 2.3 in the E-02 group (dose 750 mg/kg b.wt) compared with 15.2 ± 2.4 in the control group of aged rats. The number of ejaculations increased from 1.1 ± 0.3 in the control-aged group to 2.6 ± 0.4 in the E-01 group. The LPE of male rats was 14.2 ± 1.8 min in the control-aged group. The LPE of the aged group was reduced to 9.8 ± 1.5 min, 9.8 ± 1.6 min, and 11.4 ± 1.8 min when treated with E-01 at a dose of 300 mg/kg b.wt, and E-02 at a dose of 300 mg/kg b.wt and 750 mg/kg b.wt, respectively.

Conclusion: It was established that oral administration of lipid-based suspension of dry extract of *Epimedium koreanum* in wheat germ oil improved erectile function of aged rats.

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Keywords: *Epimedium koreanum*; Sexual behavior; Rats; Wheat germ oil

1. Introduction

Aging is a complex process with multiple alterations in the physiological structure and functional responses of the organism. In recent years, epidemiologic studies have suggested that aging is an independent predictor of erectile dysfunction, irrespective of comorbid disease states, such as cardiovascular diseases, that are conventionally understood to be major factors for this condition. Longitudinal analysis of an aging study showed that the incidence rate of erectile dysfunction rises from

1.2% per year for men aged between 40 years and 49 years, to 4.6% for men aged between 60 years and 69 years (Johannes et al., 2000). Since the elderly population is growing markedly, the number of men with erectile dysfunction is estimated to be 322 million by 2025 (Aytac et al., 1999). In recent years, there has been a renewed interest in the search for novel herbs active against erectile dysfunction (Adimoelja, 2000; Drewes et al., 2003; Kaphle et al., 2006; Lines and Ono, 2006; Abdulwaheb et al., 2007).

More than 130 different compounds were identified from over 16 species of the *Epimedium* genus (Berberidaceae). Eight of these species have been used in traditional Chinese medicines over centuries to treat a wide range of diseases (Wu et al., 2003; Shen et al., 2007). *Epimedium* Herba, the aerial part of

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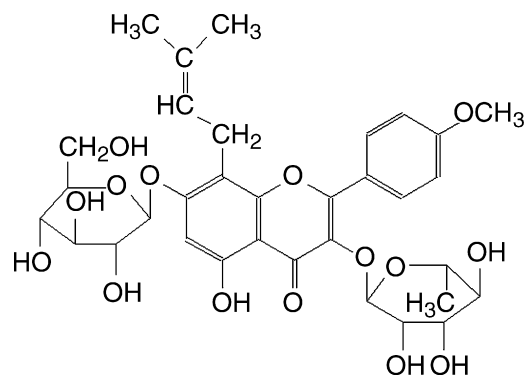


Fig. 1. Structure of icariin.

Epimedium koreanum Nakai (Berberidaceae), is widely used in traditional Chinese and Korean herbal medicine (Kee, 2000; Korea FDA, 2002; Oh et al., 2004). This medicinal herb is believed to be a potent enhancer of erectile function. Icariin (Fig. 1.) has been determined to be the main active component of *Epimedium Herba* (Ito et al., 1988). It possesses many biological effects, such as improving cardiovascular function, hormone regulation, immunological function modulation, and anti-tumor activity (He et al., 1995; Wu et al., 2003; Liu et al., 2005a,b).

Wheat germ oil in comparison with known plant fatty oils is enriched with vitamin E (up to 12,000 ppm), and the α -tocopherols, the most active forms, prevail. Wheat germ oil contains free fatty acids, which include erucic acid, myristic acid, gondoic acid, palmitic acid, stearic acid, oleic acid, linoleic acid, linolenic acid, and traces of arachidonic acid. Other minor constituents are sterols (12,000–16,400 ppm), vitamins, and lecithin.

Wheat germ oil has antiatherosclerotic and cardioprotective properties, reduces the level of cholesterol in blood and liver, and regulates hormonal balance (Shikov et al., 2004). A positive effect of wheat germ oil on androgenic status and copulative function has been reported (Pushkar et al., 2005).

This study supports the traditional use of extracts from *Epimedium* species in erectile dysfunction. The purpose of the present work was to determine the effect of oral administration of a lipid-based suspension of *Epimedium koreanum* extract standardized by icariin content in wheat germ oil on sexual behavior in male rats.

2. Materials and methods

2.1. Drugs and chemicals

The following drugs and chemicals were used: commercially available water–alcohol soluble extracts of *Epimedium koreanum* Herba were obtained from DIOD Ltd. (Moscow, Russia); estradiol benzoate (Mesalin[®]), Intervet, The Netherlands; progesterone, PharmaDon Company, Russia; olive oil, consul, Spain; wheat germ oil (Viardo[®]), DIOD Ltd., Russia; lecithin, Central Soya European Lecithins GmbH & Co. KG, Germany; bee wax, Pharmamedical, Russia.

2.2. Drug preparation

Preparation with *Epimedium Herba* in traditional Chinese and Korean medicine is administered orally; therefore, the *Epimedium* extract was suspended in wheat germ oil using lecithin and bee wax for oral administration. Two lipid-based suspensions of dry extract of *Epimedium koreanum*, E-01 (3% of icariin in extract) and E-02 (20% of icariin in extract) were used in experiments. The icariin content was determined by high-performance liquid chromatography (Pozharitskaya et al., 2007). HPLC analyses were performed at room temperature using the Waters HPLC system (USA) with two 515 series isocratic pumps, automated gradient controller and UV-detector of 484 series. Separation was achieved using an Luna C₁₈ (4.6 mm i.d. × 150 mm, 5 μ m) column (Phenomenex, USA) and SecurityGuard pre-column (20 mm) with the same sorbent (Phenomenex, USA) operating at room temperature with a flow rate of 1.0 mL/min. Data system consisted of Multi-Chrom v. 1.5x (Ampersand, Moscow, Russia) PC software. Linear gradient elution from 20% of acetonitrile in 0.03% solution of trifluoroacetic acid in water to 100% of acetonitrile in 40 min was used. UV-detection was at 270 nm, injection volume was 20 μ L. Each determination was carried out in triplicate. Acetonitrile of HPLC grade was purchased from Vekton (St.-Petersburg, Russia); trifluoroacetic acid was obtained from Sigma–Aldrich (Seelze, Germany). Before use, the solvents were filtered through a 0.45 μ m Millipore membrane (Millipore, Saint-Quentin, Yvelines, France) after sonication for 15 min. The reference substance of icariin (purity > 99%) was purchased from Chromadex (Santa Ana, CA, USA).

Estradiol benzoate and progesterone were used as oil drugs for subcutaneous injection.

2.3. Animals

Male young (14–16 weeks, weighing 250–300 g) and aged (20–22 months old, weighing 300–350 g) and virgin female (14–16 weeks, weighing 200–250 g) albino rats of the Wistar strain were obtained from the Pavlov Medical University of the Russian Academy of Sciences (St.-Petersburg, Russia). The rats were housed singly in standard cages and maintained under standard laboratory conditions (temperature 19–25 °C, relative humidity 50–70%, 12 h light/12 h dark cycle) with free access to solid pellet (Volosovo, Russia) diet and water *ad libitum* throughout the study, except during the experiment. The experimental procedures were in accordance with the directive No. 267 from 19.06.2003 “About the statement of regulation of laboratory practice” of the Ministry of Health of the Russian Federation.

2.4. Mating behavior test

The male rats were randomly assigned to one of the following groups ($N = 10$ per group): control group of young rats 300 mg/kg body weight (b.wt), which received of olive oil; control group of aged rats, which received 300 mg/kg b.wt of olive oil, groups 3–5 were aged rats, which received lipid-based suspensions at doses of 300 mg/kg b.wt (E-01 and E-02) and

750 mg/kg b.wt (E-02), respectively, daily for 10 days. The route of administration for all treatment was oral by intragastric tube. The male rats sexual behavior tests were conducted on the 11th day after commencement of the treatment of the male animals and 2 h after the onset of darkness. Single male rats were placed in a rectangular observation cage and allowed to acclimatize to the test chamber for 20 min and exposed to red light. Then a sexually receptive female (stimulus) was presented to the male by dropping it gently into the cage. The female rats were brought into estrus with a single subcutaneous dose of estradiol benzoate and progesterone (Zheng et al., 2000). They were administered a solution of estradiol benzoate orally at the dose of 200 mg/kg b.wt 48 h before the pairing plus progesterone injected subcutaneously at a dose of 2.5 mg/kg b.wt 6 h before the experiment. The receptivity of the female animals was confirmed before the test by exposing them to male animals, other than the control, test and standard animals. Mating behavior test was measured according to the Guideline on Experimental (Pre-clinical) Studying of New Pharmacological Substances (2005) and the standard procedure by Agmo (1997). Tests were terminated immediately after the first post-ejaculatory intromission; or if intromission did not occur within 15 min; or if ejaculatory latency exceeded 30 min; or in the case where post-ejaculatory interval exceeded 15 min. The occurrence of events and phases of mating were called out to be recorded on video cassette as soon as they appeared. Their disappearance was also called out and recorded. Later, the frequencies and phases, the number of complete intromissions, and the latent period of ejaculation were determined from cassette transcriptions.

The next morning, a vaginal smear from each female rat was examined under a light microscope for the presence of sperm. The number of ejaculations was recorded.

2.5. Statistical analysis

Pairwise statistical comparisons between the control and treated groups were done with Student's *t*-test. Mean differences were considered statistically significant if $P < 0.05$. The data are presented as mean $\pm$ S.D.

3. Results

The *Epimedium* extract formulations E-01 and E-02 were administered orally to the normal male rat experimental groups at doses of 300 mg/kg b.wt (E-01 and E-02) and 750 mg/kg b.wt (E-02) for 10 days. The control groups received olive oil for 10 days at a dose of 300 mg/kg b.wt. The number of complete intromissions during the 30 min study period (after 10 days feeding) increased to 23.3 ± 2.6 ($P < 0.05$) in the E-01 and E-02 experimental groups (dose 300 mg/kg b.wt) and to 20.1 ± 2.3 ($P < 0.05$) in the E-02 experimental group (dose 750 mg/kg b.wt) compared with 15.2 ± 2.4 in the control group of aged rats (Fig. 2).

In the assessment of 1 day of mating, the oral administration of E-01 at a dose of 300 mg/kg b.wt increased the number of ejaculations from 1.1 ± 0.3 in the control group of aged rats to 2.6 ± 0.4 ($P < 0.05$) in the E-01 experimental group (Fig. 3).

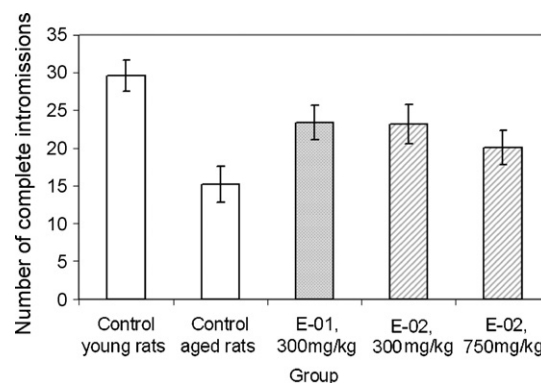


Fig. 2. The effect of lipid-based suspension of dry extracts of *Epimedium koreanum* and wheat germ oil (E-01 and E-02) on the number of complete intromissions in normal male rats during 30 min. Values are presented as the mean $\pm$ S.D.; $N = 10$. $P < 0.05$ for control-aged rats by pairwise comparison with the control group of young rats; $P < 0.05$ for E-01 and E-02 data by pairwise comparison with the control group of aged rats. The control groups received olive oil. E-01 and E-02 were administered orally to normal male rats for 10 days.

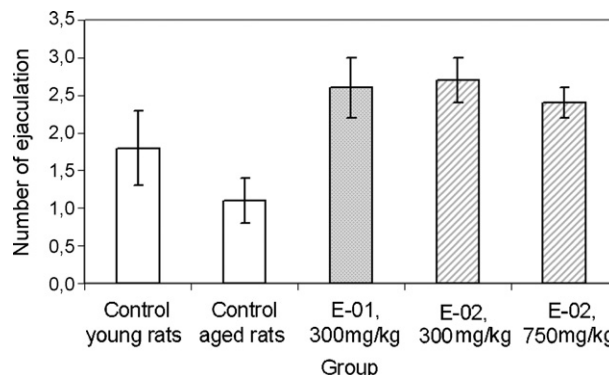


Fig. 3. The effect of lipid-based suspension of dry extracts of *Epimedium koreanum* and wheat germ oil, E-01 and E-02, on the number of ejaculations. Values are presented as the mean $\pm$ S.D.; $N = 10$. $P < 0.05$ for E-01 and E-02 by pairwise comparison with the control group of aged rats.

The latent period of ejaculation (LPE) in the control group of young rats was 11.0 ± 1.9 min. The LPE in the control group of aged rats was 14.2 ± 1.8 min. As predicted, the aged rats demonstrated the weakest sexual ability, as exhibited by the longest LPE values. The LPE in the aged group was reduced to 9.8 ± 1.5 min, 9.8 ± 1.6 min, and 11.4 ± 1.8 min when treated with E-01 at a dose of 300 mg/kg b.wt, and E-02 at dose of 300 mg/kg b.wt and 750 mg/kg b.wt, respectively (Table 1).

Table 1
The latent period of ejaculation (LPE) in rats

Group ($N = 10$)	Dose (mg/kg)	LPE (min)
Control young rats		11.0 ± 1.9
Control aged rats		14.2 ± 1.8
E-01	300	9.8 ± 1.5
E-02	300	9.8 ± 1.6
E-02	750	11.4 ± 1.8

4. Discussion

Clinical research of novel natural substances has made particularly rewarding progress in many important fields such as anticancer drugs, antirheumatic drugs and others. Natural product research can often be guided by ethnopharmacological knowledge, and it can give substantial contribution to drug innovation by providing novel chemical structures and/or mechanisms of actions (Harvey, 1999).

Sexual incompetence, including erectile dysfunction, is a problem of increasing concern (NIH Consensus Development Panel on Impotence, 1993) and is not exclusive to humans. Among male rats, a variable percentage of subjects is sexually impotent, i.e. is unable to conclude with ejaculation, or even to initiate sexual intercourse with a receptive female, thus they can be considered a good model to study the sexual impotence (Agmo, 1997).

Traditionally *Epimedium* Herba has been used as a tonic drug in traditional Chinese medicine for centuries. This medicinal herb is believed to be a potent enhancer of erectile function. Extracts of *Epimedium* spp., which is rich in prenylated flavonoids, has a vasodilating effects mediated by NO release, it had the extract performing a vasokinetic activity on the arteries and arterioles (Liu et al., 2005a,b; Chiu et al., 2006). It was established that icariin is a cGMP-specific PDE5 inhibitor that may be developed into an oral effective agent for the treatment of ED (Xin et al., 2003). Ning et al. (2006) have shown, that icariin enhance cGMP levels in SNP-treated cavernous smooth muscle cells.

Icariin, poorly water soluble prenylated flavonoid, is the major compound providing biological effects of *Epimedium koreanum* whose effects are caused by selective inhibiting influence on phosphodiesterase-5 (PDE5). The rational medicinal form for *per os* administration for *Epimedium koreanum* dry extract is lipid-based suspension which can be used as softgel capsules. The benefit of this form consists in a big total surface that provides the best therapeutic effect and prolonged action. Lipid-based delivery systems are becoming increasingly popular as carriers of drugs due to their ability to overcome barriers to oral absorption. Wheat germ oil was chosen as lipid base for *Epimedium koreanum* extract.

The data, presented in Figs. 2 and 3, and Table 1 confirmed that the aged male rat is an acceptable model of erectile dysfunction. The decrease of the number of complete intromissions of 49% and an increase of LPE of 29% was observed in control-aged rats.

The greatest changes were observed for parameters such as the number of complete intromissions and the number of sperm-positive females. For *Epimedium* formulation E-02, a decrease of unsuccessful intromissions was observed even below the level of control rats (61–66%). Groups E-01 and E-02 at a dose of 300 mg/kg b.wt were most effective for the number of complete intromissions and the number of sperm-positive females, and the number of complete intromissions was increased by 53–54% in comparison with control animals, and practically achieved the level of young rats. The number of sperm-positive females was increased in comparison with control young rats by 44% and

50%. The evaluation of vaginal smears confirmed estrus, ante estrus and post-estrus.

Prenylflavonoid icariin is the major component providing the pharmacological effects of *Epimedium*. Xin et al. (2003) studied the mechanism of therapeutic action of icariin on erectile dysfunction and established its dose-dependent selective inhibitory effect on phosphodiesterase-5 (PDE5). Oral treatment with icariin (>98.6% purity) for 4 weeks potentially improves erectile function. This effect is correlated with an increase in the percentage of smooth muscle and the expression of certain NO synthase isoforms in the corpus cavernosum of castrated rats. These results suggest that icariin may have a therapeutic effect on erectile dysfunction (Liu et al., 2005b). Icariin was inhibitory to all three PDE5 isoforms and with similar IC₅₀ values, which were approximately three times greater than those for zaprinast. Icariin was able to enhance the levels of cyclic guanosine monophosphate in sodium nitroprusside-treated cavernous smooth muscle cells (Ning et al., 2006).

The results of our experiments with two compositions of *Epimedium* extract in wheat germ oil, E-01 and E-02, on the mounting behavior in aged rats showed that *Epimedium* products enhance the sexual libido and potency in male rats significantly.

The results of the *in vivo* study showed that the fixed composition of *Epimedium* extracts in wheat germ oil, E-01 and E-02, were effective in improving the erectile function of aged rats. As demonstrated by the decrease in LPE, the erectile function of the aged group treated with the extract was better than that of the control-aged group, and was almost identical with that of control young rats. It is probably that decrease of LPE in treated rats is associated with the higher concentration of icariin and fatty acids and sterols, a group of biologically active components of *Epimedium* extract and wheat germ oil.

5. Conclusions

Two lipid-based suspensions of dry extract of *Epimedium* with wheat germ oil, E-01 and E-02, enhanced sexual potency in normal young rats and in aged rats significantly. The number of complete intromissions during a 30 min period in normal male rats administered E-01 and E-02 were 1.5 times higher, on the average, than that of the normal rats in the control-aged group. Moreover, a single oral administration of E-01 showed that the number of ejaculations in the experimental group was 2.5 times higher than in the control-aged group. Finally, E-01 and E-02 improved erectile function in rats with erectile dysfunction. The LPE of aged rats that were administered different doses of E-01 and E-02 was reduced to that of normal young rats. The present study reveals for the first time an aphrodisiac effect of dry extract of *Epimedium koreanum* in wheat germ oil on rat's model.

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